Dual Compressed Decorrelation Regularizer (DCD)

Make Features Independent, Make Models Dependable

CSC 591 (007) Deep Learning Beyond Accuracy

Instructor: Jung-Eun Kim

Ajith Kanumuri, Mrudani Hada, Parth Parikh

Overview

1. Deep neural networks often learn overlapping filters that are actually redundant features. These features waste model capacity and can impair generalization.

2. Reducing redundancy by encouraging feature diversity may lead to richer, more efficient internal representations, ultimately boosting both accuracy and robustness.

3. **Goal:** Design a decorrelation regularizer that encourages model to reduce redundant features.

4. **Models:** *VGG-16 & EfficientNet* || **Dataset:** *CIFAR-10 & CIFAR-100*

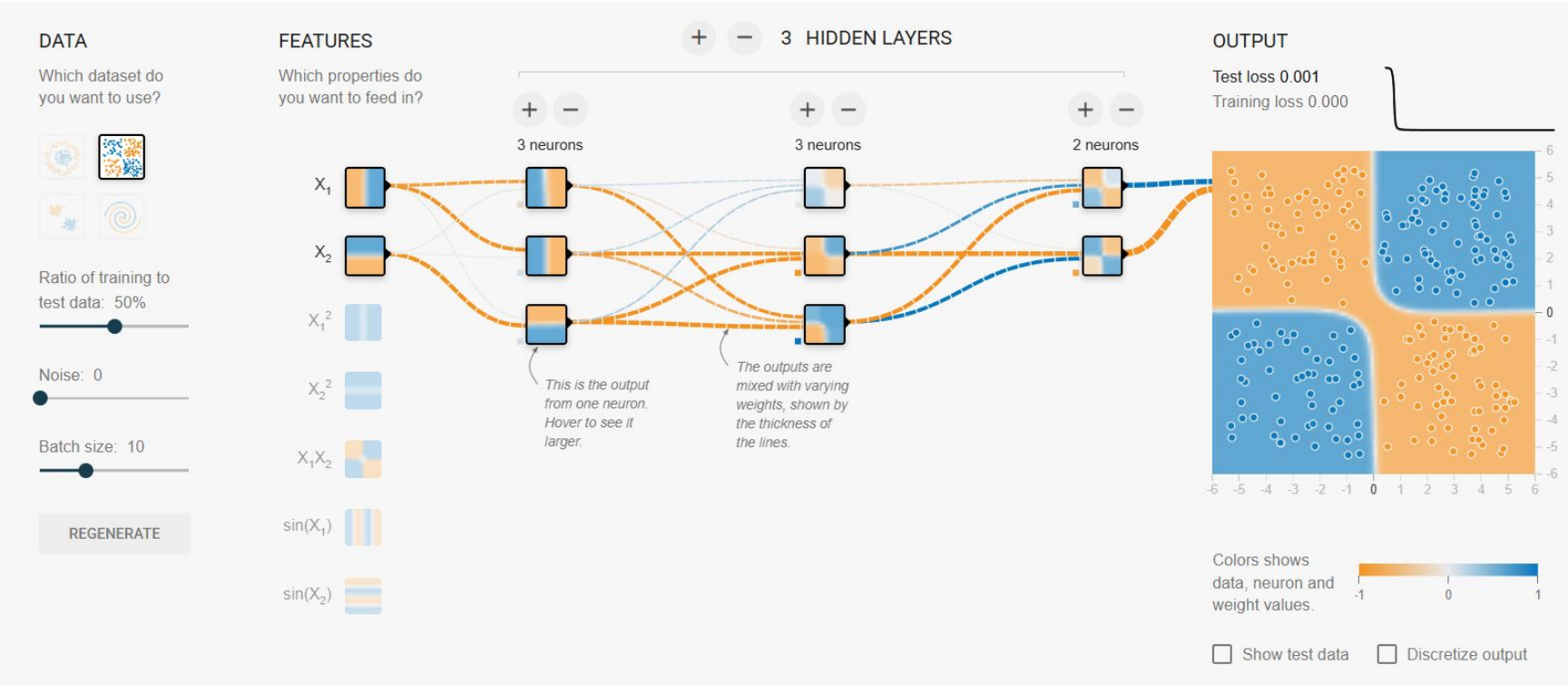
Limitations of Existing Work

1. DeCov[1]:
   - Computationally expensive due to full covariance matrix computation
   - Ineffective for transfer learning
   - Outdated compared to orthogonality-based regularizers

2. Hard orthogonality[2]:
   - Loss of representational flexibility, leading to less accuracy
   - Computationally expensive due to SVD-like computation

[1] https://arxiv.org/pdf/1511.06068
[2] https://proceedings.neurips.cc/paper_files/paper/2018/file/bf424cb7b0dea050a42b9739eb261a3a-Paper.pdf

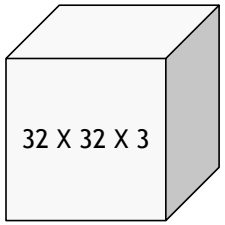
Example of Redundancy in an NN



Method

Activation Decorrelation Regularization (La)

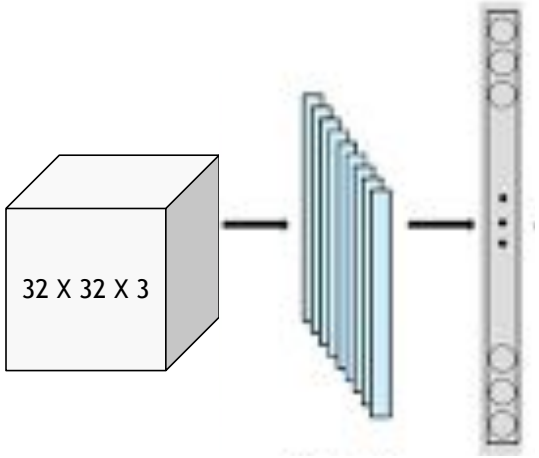
Objective: Encourage diversity and independence among hidden activations, reducing redundancy in learned representations.



Input Image

Activation Decorrelation Regularization (La)

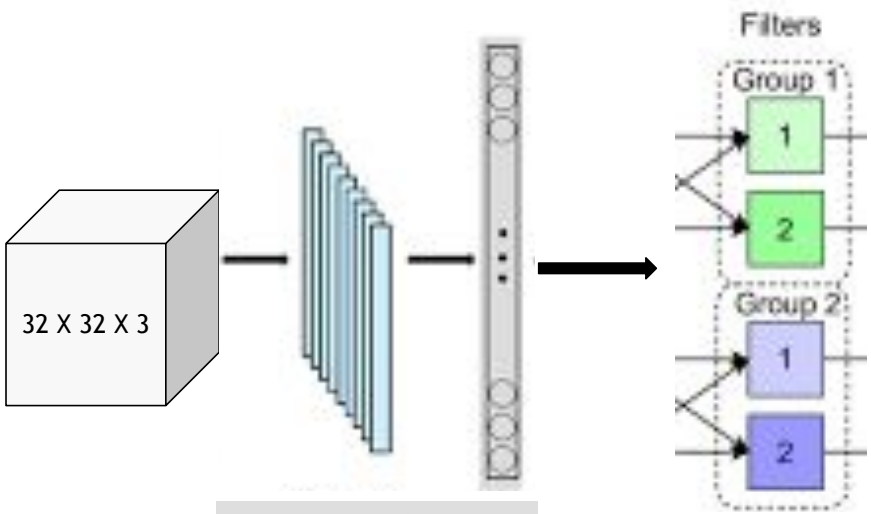
Objective: Encourage diversity and independence among hidden activations, reducing redundancy in learned representations.



$$A \in \mathbb{R}^{B \times C \times H \times W}$$

Activation Decorrelation Regularization (La)

Objective: Encourage diversity and independence among hidden activations, reducing redundancy in learned representations.

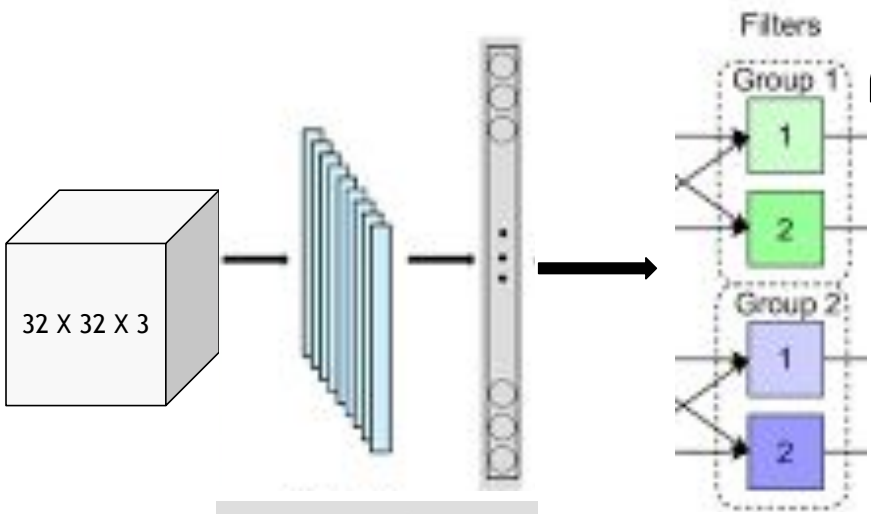


$A \in \mathbb{R}^{B \times C \times H \times W}$

$X_g \in \mathbb{R}^{N \times c_g}$

Activation Decorrelation Regularization (La)

Objective: Encourage diversity and independence among hidden activations, reducing redundancy in learned representations.



$A \in \mathbb{R}^{B \times C \times H \times W}$

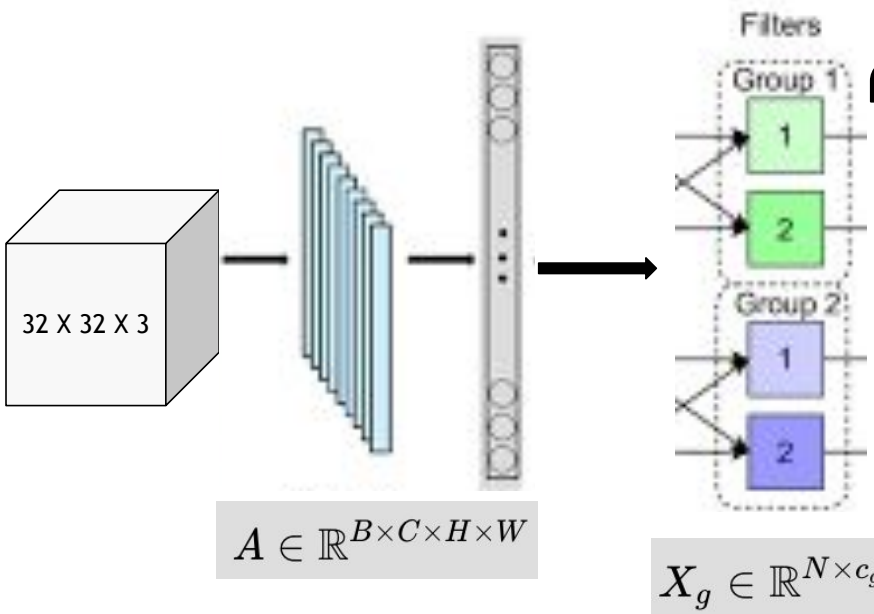
$X_g \in \mathbb{R}^{N \times c_g}$

$$Z_g = X_g S_g$$

$$S_g \in \mathbb{R}^{c_g \times k}, \quad S_g \sim \mathcal{N}(0, 1/\sqrt{k})$$

Activation Decorrelation Regularization (La)

Objective: Encourage diversity and independence among hidden activations, reducing redundancy in learned representations.



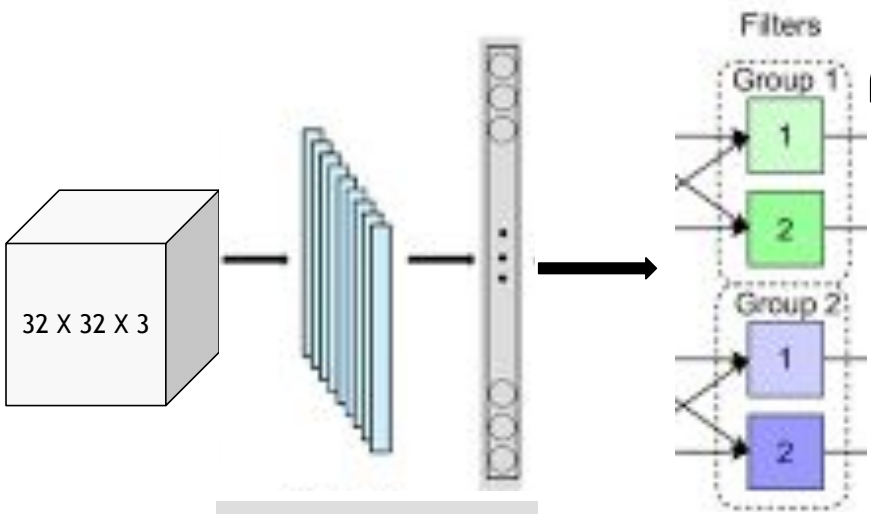
$$Z_g = X_g S_g$$

$$S_g \in \mathbb{R}^{c_g \times k}, \quad S_g \sim \mathcal{N}(0, 1/\sqrt{k})$$

$$\Sigma_g = \frac{1}{N-1}(Z_g - \bar{Z}_g)^\top (Z_g - \bar{Z}_g)$$

Activation Decorrelation Regularization (La)

Objective: Encourage diversity and independence among hidden activations, reducing redundancy in learned representations.



$A \in \mathbb{R}^{B \times C \times H \times W}$

$X_g \in \mathbb{R}^{N \times c_g}$

$$Z_g = X_g S_g$$

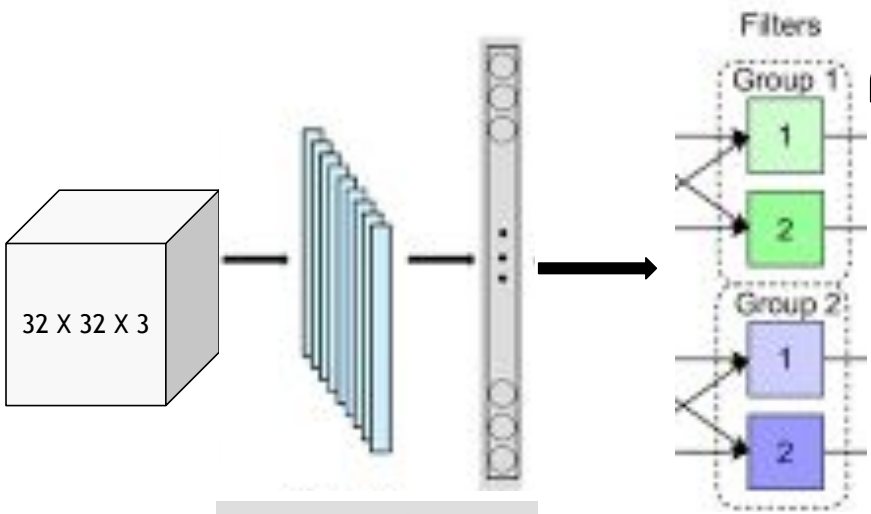
$$S_g \in \mathbb{R}^{c_g \times k}, \quad S_g \sim \mathcal{N}(0, 1/\sqrt{k})$$

$$\Sigma_g = \frac{1}{N-1}(Z_g - \bar{Z}_g)^\top (Z_g - \bar{Z}_g)$$

$$L_a = \sum_{g=1}^{G} \|\Sigma_g - \mathrm{diag}(\Sigma_g)\|_F^2$$

Activation Decorrelation Regularization (La)

Objective: Encourage diversity and independence among hidden activations, reducing redundancy in learned representations.



$A \in \mathbb{R}^{B \times C \times H \times W}$

$X_g \in \mathbb{R}^{N \times c_g}$

$$Z_g = X_g S_g$$

$$S_g \in \mathbb{R}^{c_g \times k}, \quad S_g \sim \mathcal{N}(0, 1/\sqrt{k})$$

$$\Sigma_g = \frac{1}{N-1}(Z_g - \bar{Z}_g)^\top (Z_g - \bar{Z}_g)$$

$$L_a = \sum_{g=1}^{G} \|\Sigma_g - \mathrm{diag}(\Sigma_g)\|_F^2$$

Optionally, EMA: $\Sigma_g^{(t)} = \beta \Sigma_g^{(t-1)} + (1-\beta)\Sigma_g$

Why Gaussian Sketching?

1. **Motivation for grouping:** Full DeCov needs $O(C^2)$ covariance per layer; slow & memory-heavy.
2. **What is it?** A linear dimensionality reduction technique.
3. **What does it do?** Projects a higher dimensional matrix into lower dimensional subspace using a random matrix drawn from *Gaussian Distribution*.
4. **Why does it work?** *Johnson-Lindenstrauss Lemma* shows that the geometry of data is approximately preserved after the projection.
5. So, we get DeCov-like effect with much lower cost.

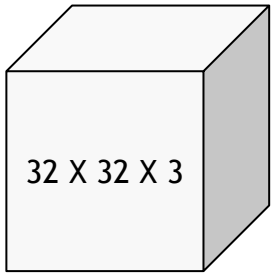
$$Z_g = X_g S_g$$

$$S_g \in \mathbb{R}^{c_g \times k}, \quad S_g \sim \mathcal{N}(0, 1/\sqrt{k})$$

$$(1 - \varepsilon)\, ||x_1 - x_2||^2 \leq ||Gx_1 - Gx_2||^2 \leq (1 + \varepsilon)\, ||x_1 - x_2||^2$$

Soft Orthogonality Regularization (Lw)

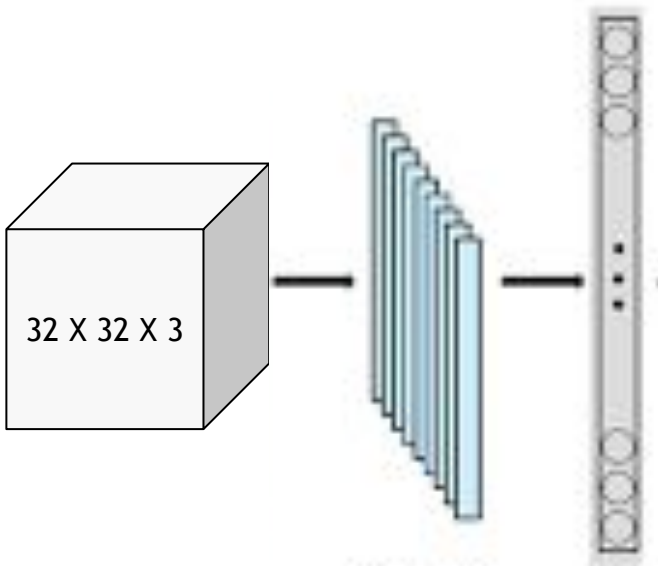
Objective: Promote near-orthogonal filters within convolutional layers to improve representation diversity and stability.



Input Image

Soft Orthogonality Regularization (Lw)

Objective: Promote near-orthogonal filters within convolutional layers to improve representation diversity and stability.



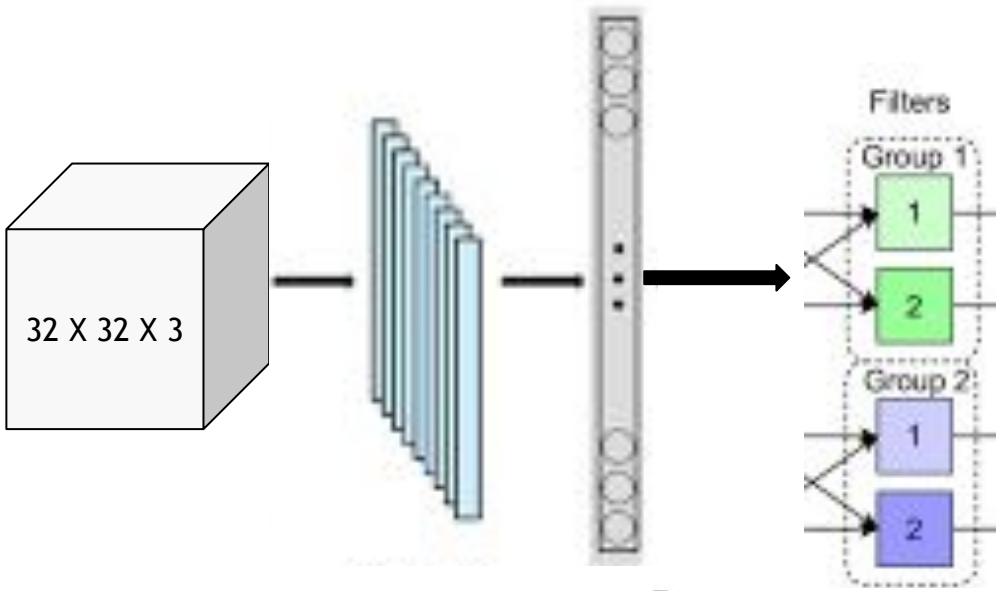
$R = C_{in} \times K_h \times K_w$

$W_f = C_{out} \text{ X } R$

$$W_f = \begin{bmatrix} — w_1^T — \\ — w_2^T — \\ \vdots \\ — w_{C_{\text{out}}}^T — \end{bmatrix}$$

Soft Orthogonality Regularization (Lw)

Objective: Promote near-orthogonal filters within convolutional layers to improve representation diversity and stability.



$R = C_{in} \times K_h \times K_w$

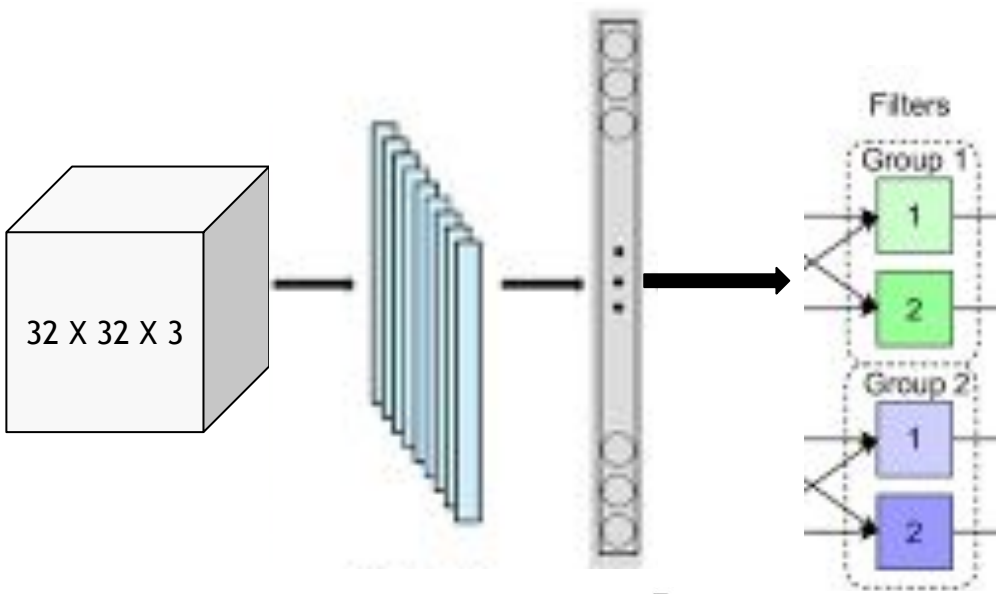
$W_f = C_{out} \times R$

$$W_f = \begin{bmatrix} — w_1^T — \\ — w_2^T — \\ \vdots \\ — w_{C_{\text{out}}}^T — \end{bmatrix}$$

$$W_g \in \mathbb{R}^{c_g \times R}$$

Soft Orthogonality Regularization (Lw)

Objective: Promote near-orthogonal filters within convolutional layers to improve representation diversity and stability.



$$W_{g,\text{proj}} = W_g P_g^\top$$

$$P_g \in \mathbb{R}^{r \times R}, \quad P_g \sim \mathcal{N}(0, 1/\sqrt{r})$$

$R = C_{in} \times K_h \times K_w$

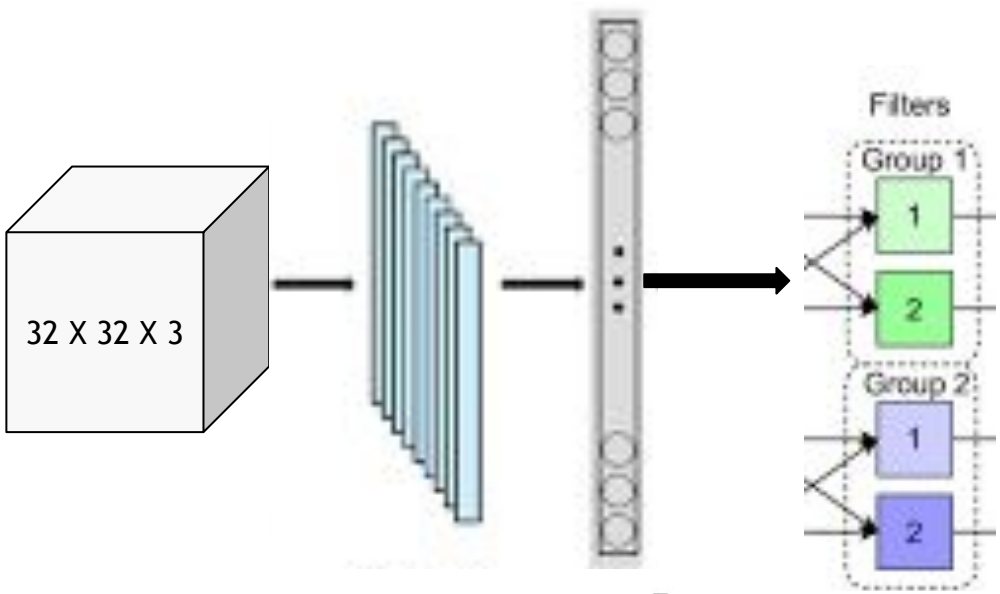
$W_f = C_{out} \times R$

$$W_f = \begin{bmatrix} — w_1^T — \\ — w_2^T — \\ \vdots \\ — w_{C_{\text{out}}}^T — \end{bmatrix}$$

$$W_g \in \mathbb{R}^{c_g \times R}$$

Soft Orthogonality Regularization (Lw)

Objective: Promote near-orthogonal filters within convolutional layers to improve representation diversity and stability.



$$W_{g,\text{proj}} = W_g P_g^\top$$

$$P_g \in \mathbb{R}^{r \times R}, \quad P_g \sim \mathcal{N}(0, 1/\sqrt{r})$$

$$L_w = \sum_{g=1}^{G} \| W_{g,\text{proj}} W_{g,\text{proj}}^\top - I_{c_g} \|_F^2$$

$R = C_{in} \times K_h \times K_w$

$W_f = C_{out} \text{ X } R$

$$W_f = \begin{bmatrix} — w_1^T — \\ — w_2^T — \\ \vdots \\ — w_{C_{\text{out}}}^T — \end{bmatrix}$$

$$W_g \in \mathbb{R}^{c_g \times R}$$

Why Soft Orthogonality?

Aspect	Soft Orthogonality	Hard Orthogonality
Constraint Type	Imposes an *approximate* orthogonality via a penalty term (in our case frobenius term)	Enforces *exact* orthogonality using explicit orthogonalization
Computational Cost	Low — added as a lightweight regularization during training.	High — requires explicit matrix operations per step
Optimization Flexibility	Smooth, differentiable, and integrates easily with optimization dynamics	Rigid, may interfere with gradient flow and optimization dynamics.
Numerical Stability	More stable since constraints are relaxed, avoiding singularities.	Prone to instability for large matrices or deep layers.
Practical Benefit	Encourages decorrelation and generalization while keeping learning efficient.	Guarantees perfect orthogonality but at the cost of slower and harder convergence.

Sparsity Regularization (Ls)

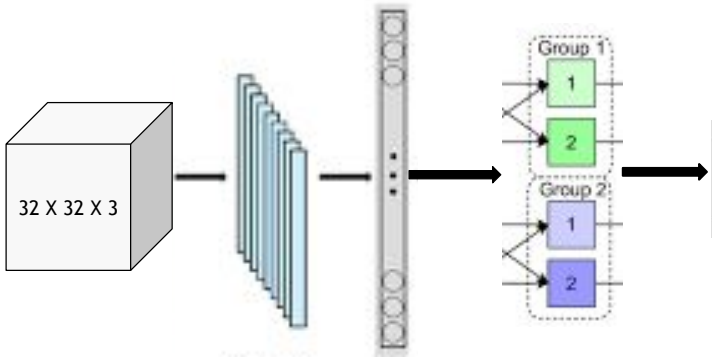
Objective: Encourage **weight sparsity** to promote **structured and unstructured pruning**.

1. Classic **L1 penalty** on weights.
2. Drives unimportant weights toward zero, enabling post-training pruning.

$$L_s = \sum_{\text{layers}} \|W\|_1 = \sum_{i,j} |W_{ij}|$$

Quantization-Aware Regularization (Lq)

Objective: Reduce sensitivity to quantization by making weights robust to rounding effects.

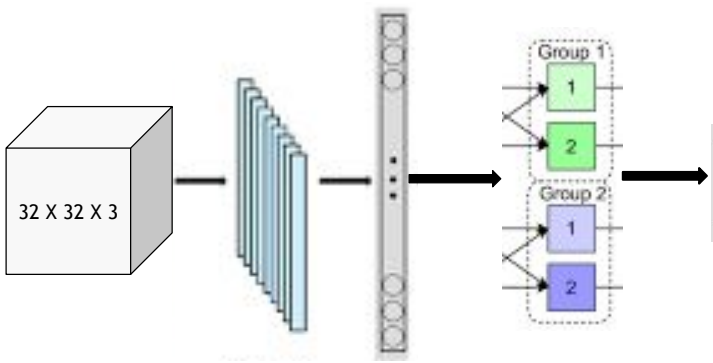


$$W_{g,\text{proj}} = W_g P_g^\top$$

$$P_g \in \mathbb{R}^{r \times R}, \quad P_g \sim \mathcal{N}(0, 1/\sqrt{r})$$

Quantization-Aware Regularization (Lq)

Objective: Reduce sensitivity to quantization by making weights robust to rounding effects.



$$W_{g,\text{proj}} = W_g P_g^\top$$

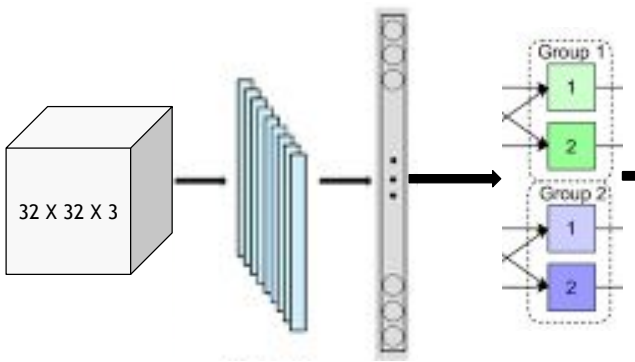
$$P_g \in \mathbb{R}^{r \times R}, \quad P_g \sim \mathcal{N}(0, 1/\sqrt{r})$$

$$s = \max(\text{mean}(|W_{g,\text{proj}}|), \epsilon)$$

$$x = W_{g,\text{proj}} / s$$

Quantization-Aware Regularization (Lq)

Objective: Reduce sensitivity to quantization by making weights robust to rounding effects.



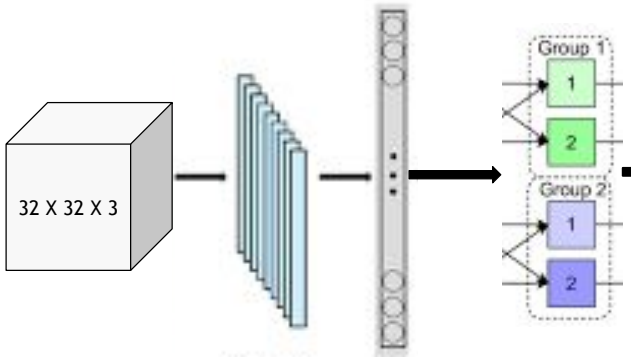
$$W_{g,\text{proj}} = W_g P_g^\top$$
$$P_g \in \mathbb{R}^{r \times R}, \quad P_g \sim \mathcal{N}(0, 1/\sqrt{r})$$

$$s = \max(\text{mean}(|W_{g,\text{proj}}|), \epsilon)$$
$$x = W_{g,\text{proj}}/s$$

$$\text{soft_round}(x) = (x - \tanh(\alpha(x - \text{round}(x))))^2$$

Quantization-Aware Regularization (Lq)

Objective: Reduce sensitivity to quantization by making weights robust to rounding effects.



$$W_{g,\text{proj}} = W_g P_g^\top$$

$$P_g \in \mathbb{R}^{r \times R}, \quad P_g \sim \mathcal{N}(0, 1/\sqrt{r})$$

$$s = \max(\text{mean}(|W_{g,\text{proj}}|), \epsilon)$$

$$x = W_{g,\text{proj}} / s$$

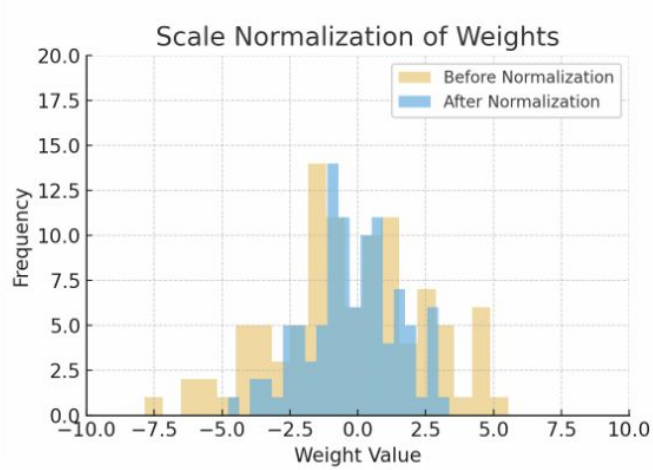
$$\text{soft_round}(x) = (x - \tanh(\alpha(x - \text{round}(x))))^2$$

$$L_q = \sum_{g=1}^{G} \sum_{i,j} \text{soft_round}(x_{ij})$$

The Need for Scale Normalization

Important: This normalization is not a standard batch or layer normalization it's a scale normalization specifically meant to stabilize the projected weights after applying the Gaussian sketch or soft orthogonality step.

1. Prevents exploding or vanishing updates by keeping average scale around 1.
2. The small constant ε avoids division by zero when weights are near zero.
3. Normalizes filter groups W_g to a consistent range before aggregation.
4. Uses mean(|W|) for efficiency — no heavy variance computation like BatchNorm.

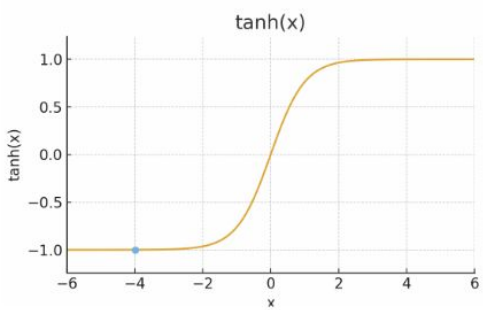


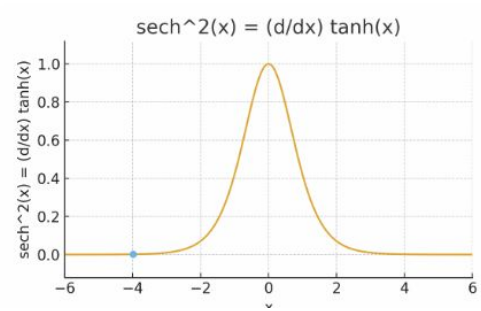
Why tanh ?

Motivation behind Soft Round: Soft rounding is chosen over hard rounding in quantization aware regularization because it provides a differentiable approximation of the rounding operation, allowing gradient flow during training. This smooth transition prevents instability and improves optimization and generalization compared to non-differentiable hard rounding.

Key reasons for choosing tanh:

1. *Bounded Output*
2. *Smooth Differentiability*
3. *Avoiding ReLU*





Final Loss function

$$\mathcal{L}_{\mathrm{DCD}} = \lambda_a L_a + \lambda_w L_w + \lambda_s L_s + \lambda_q L_q$$

where:

$\lambda_a \rightarrow$ decorrelation strength

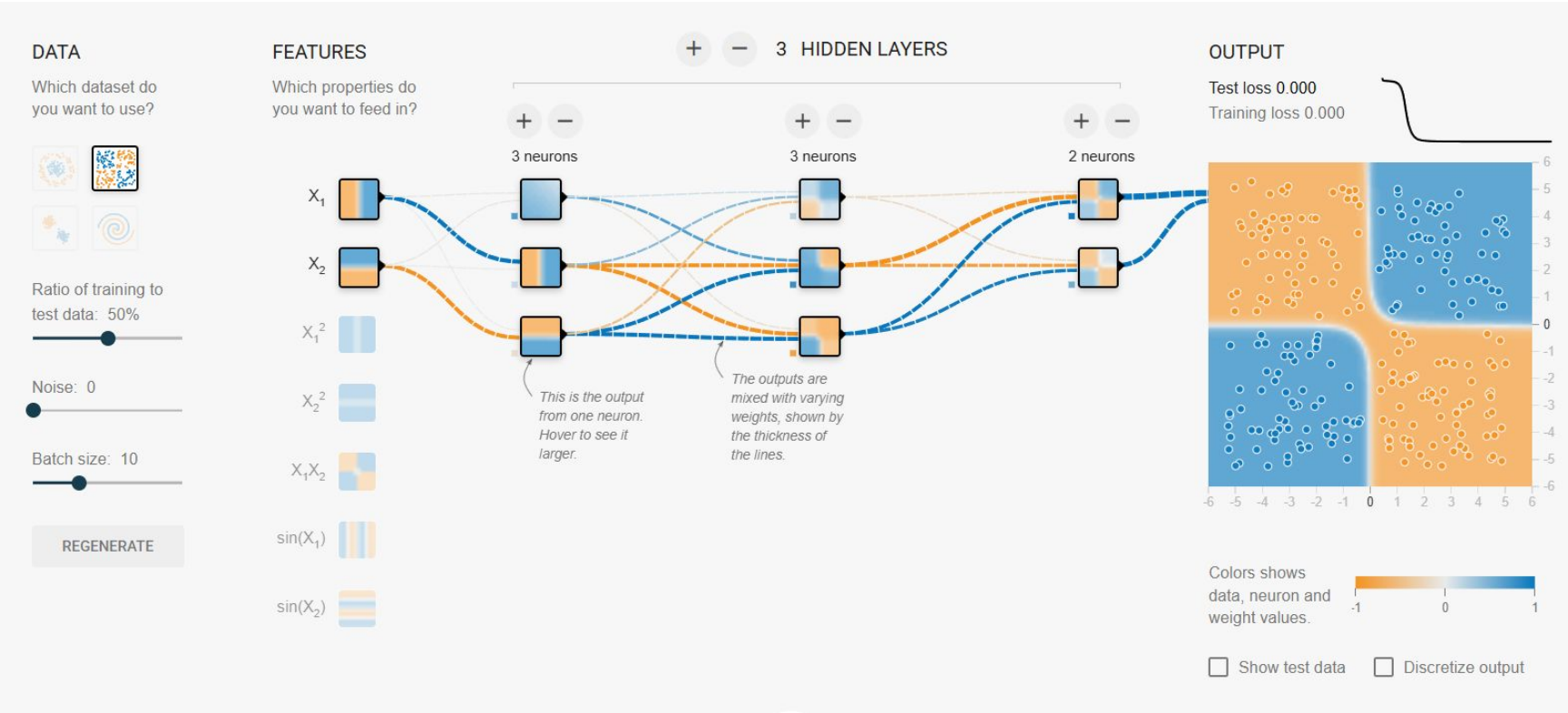
$\lambda_w \rightarrow$ orthogonality strength

$\lambda_s \rightarrow$ sparsity control

$\lambda_q \rightarrow$ quantization-awareness weight

(L_a)	Activation decorrelation loss
(L_w)	Soft orthogonality loss
(L_s)	Sparsity loss (L1)
(L_q)	Quantization-aware loss

Effect of Our Regularizer



Challenges Faced

1. **Model Selection for Redundancy Testing:** Evaluated different models; selected EfficientNet and VGG for optimal redundancy testing.

2. **Selecting the Most Impactful Layers:** Tested convolutional stages and found mid-to-deep layers showed the strongest redundancy reduction.

3. **Optimal Filter Grouping Strategy:** Balancing group size in Gaussian sketching was challenging, requiring covariance preservation with lower computation.

4. **Choosing the best Normalization Strategy:** A key challenge was stabilizing orthogonalized weights; scaled normalization was required to maintain smooth gradients.

Metrics

Our initial expectations…

We hypothesized that applying our regularizer would lead to:

1. Lower Feature Redundancy — reduced correlation between feature channels.

2. Greater Representational Diversity — filters learn distinct, specialized features.

3. Higher Test Accuracy — model captures richer and more generalizable patterns.

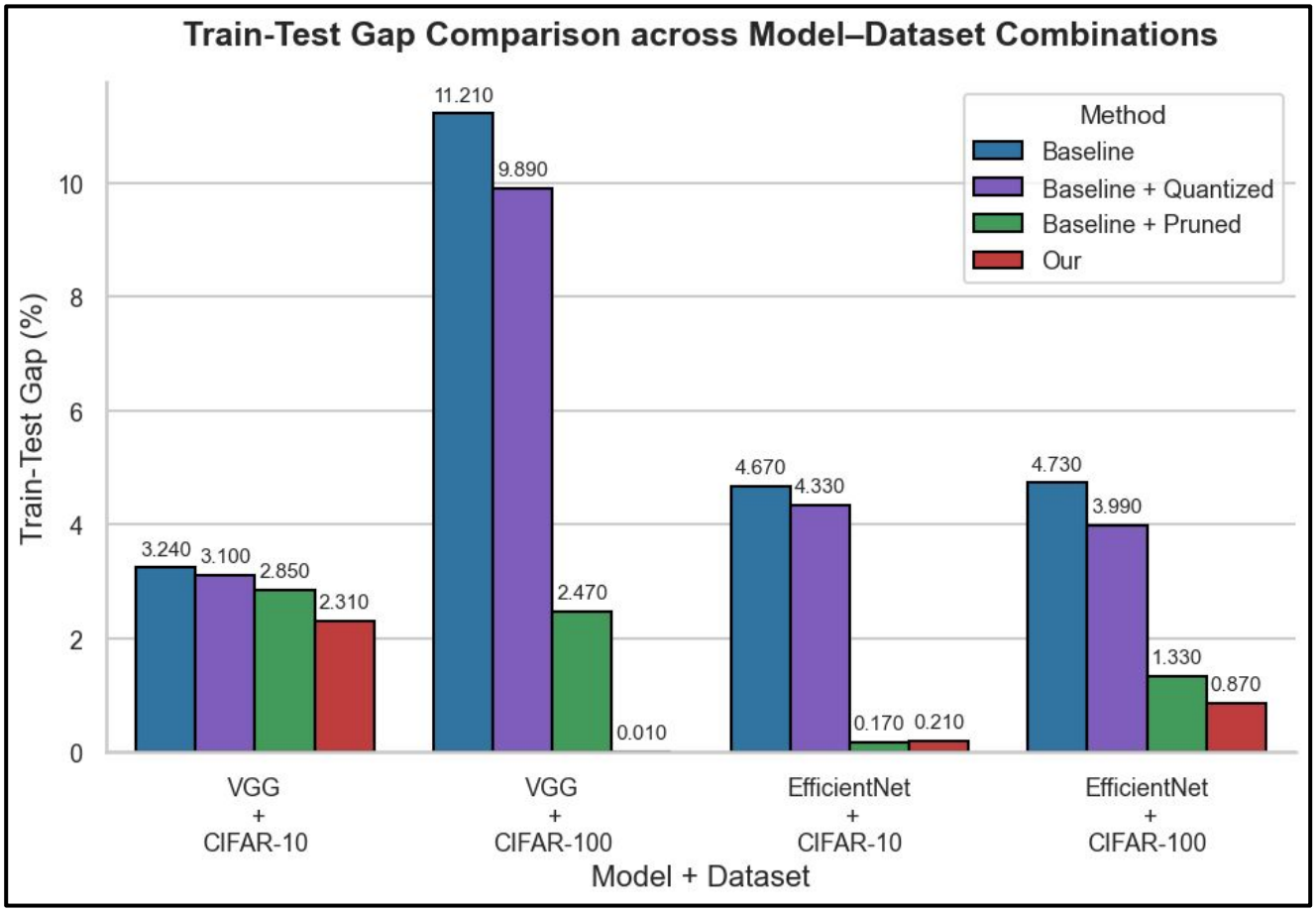
In order to achieve those…

EXPECTED PROPERTY	CHOSEN METRIC	MATHEMATICAL REPRESENTATION
Generalization	Train-Test Accuracy Gap	$\Delta_{\text{Acc}} = \lvert \text{Acc}_{\text{train}} - \text{Acc}_{\text{test}} \rvert$
Calibration /Reliability	Expected Calibration Error	$\text{ECE} = \sum_{m=1}^{M} \frac{\lvert B_m \rvert}{N} \lvert \text{acc}(B_m) - \text{conf}(B_m) \rvert$
Representation Quality/ Redundancy	Mean Off-Diagonal Covariance	$\text{MOC} = \frac{1}{d(d-1)} \sum_{i \neq j} \lvert \rho_{ij} \rvert$
Overall Performance	Top-1/Top-5 Test Accuracy	$\text{Acc}_1 = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}\left(y_i = \hat{y}_i^{(1)}\right)$ $\text{Acc}_5 = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}\left(y_i \in \text{Top-5}(\hat{y}_i)\right)$
Training Alignment	Validation Cross Entropy Loss	$\mathcal{L}_{val} = -\frac{1}{N_{val}} \sum_{i=1}^{N_{val}} \sum_{c=1}^{C} y_{i,c} \log \hat{p}_{i,c}$

Results & Analysis

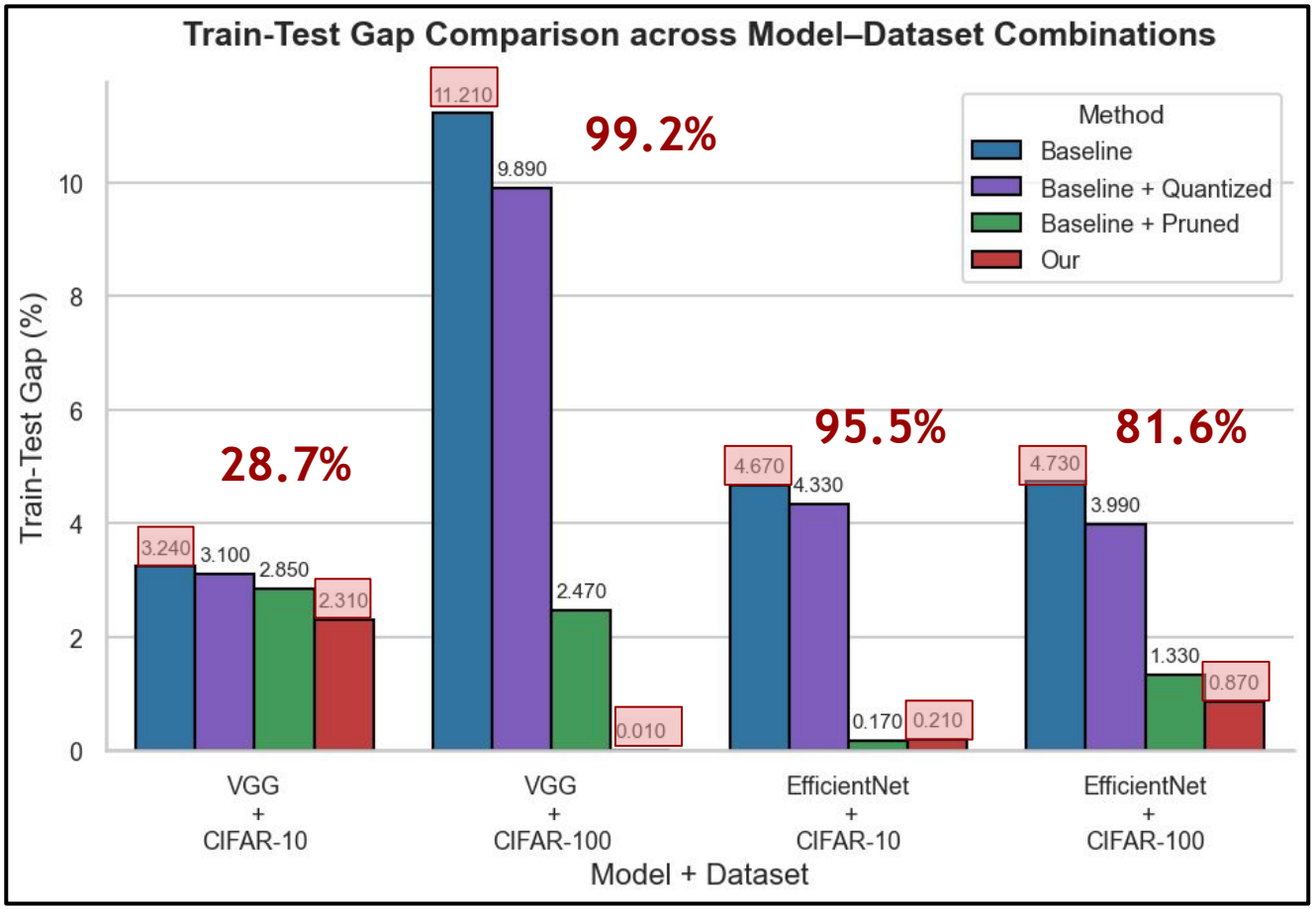
Train-Test Accuracy Gap

↓ Train-Test Gap → better generalization



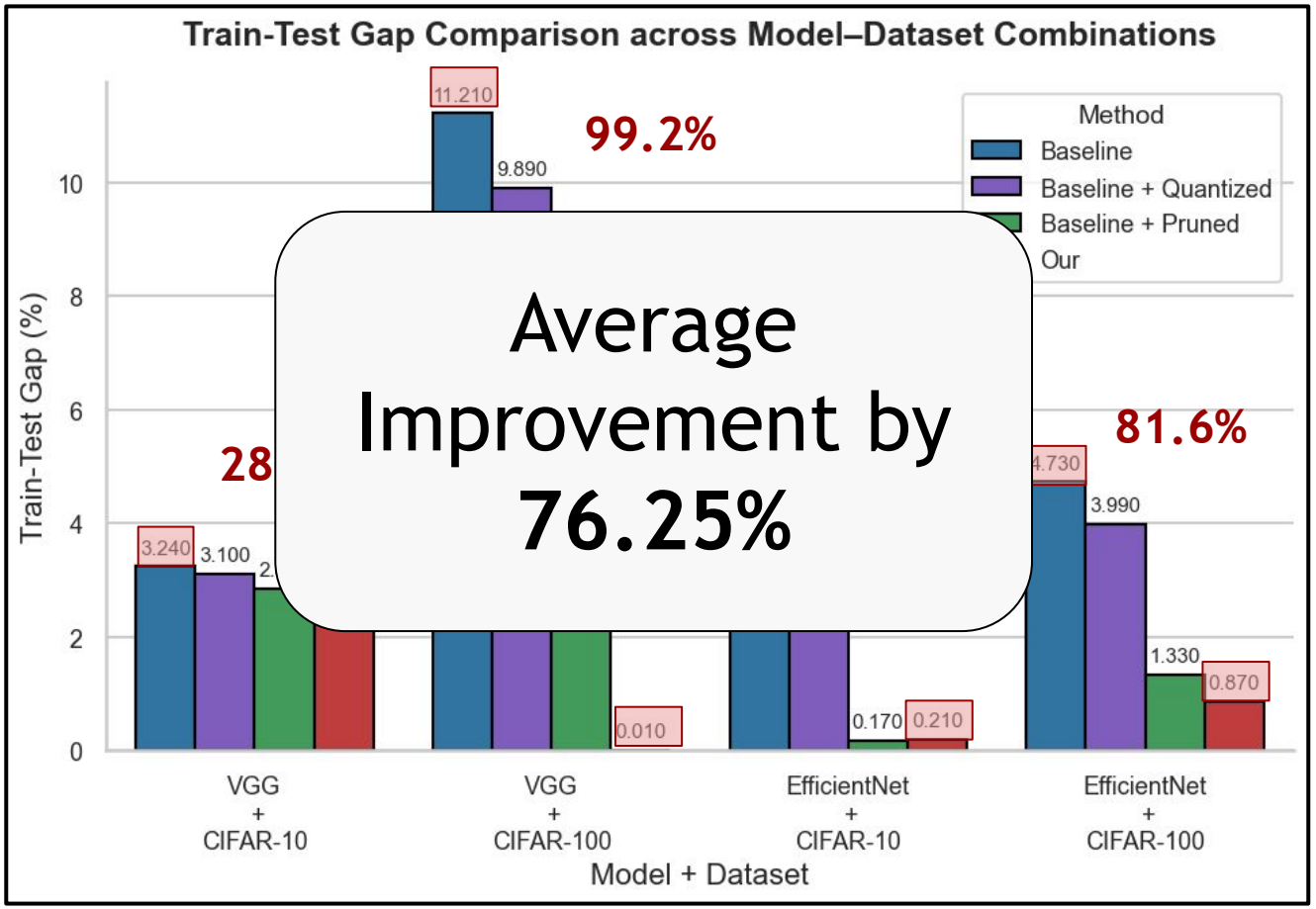
Train-Test Accuracy Gap

↓ Train-Test Gap → better generalization



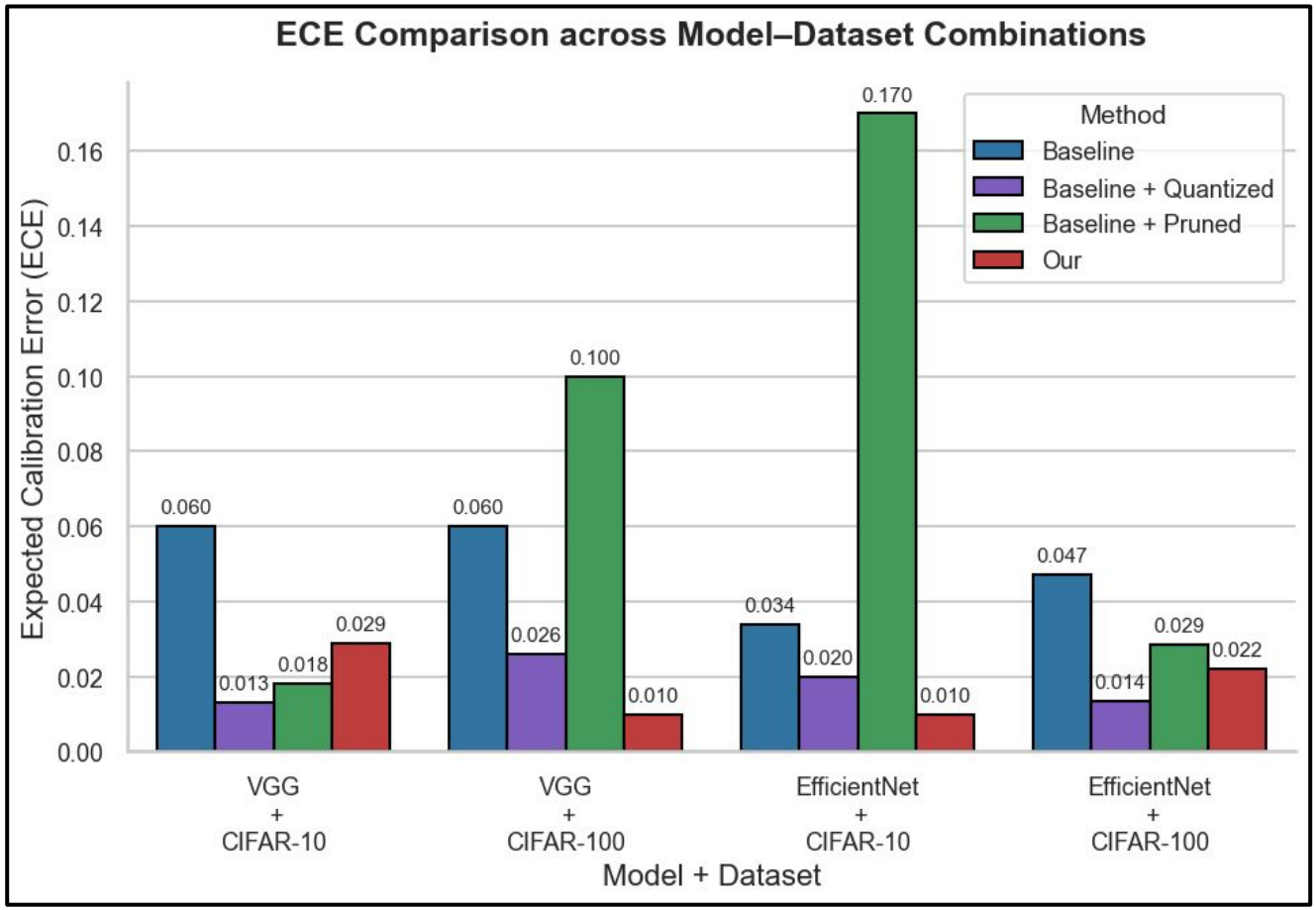
Train-Test Accuracy Gap

↓ Train-Test Gap → better generalization



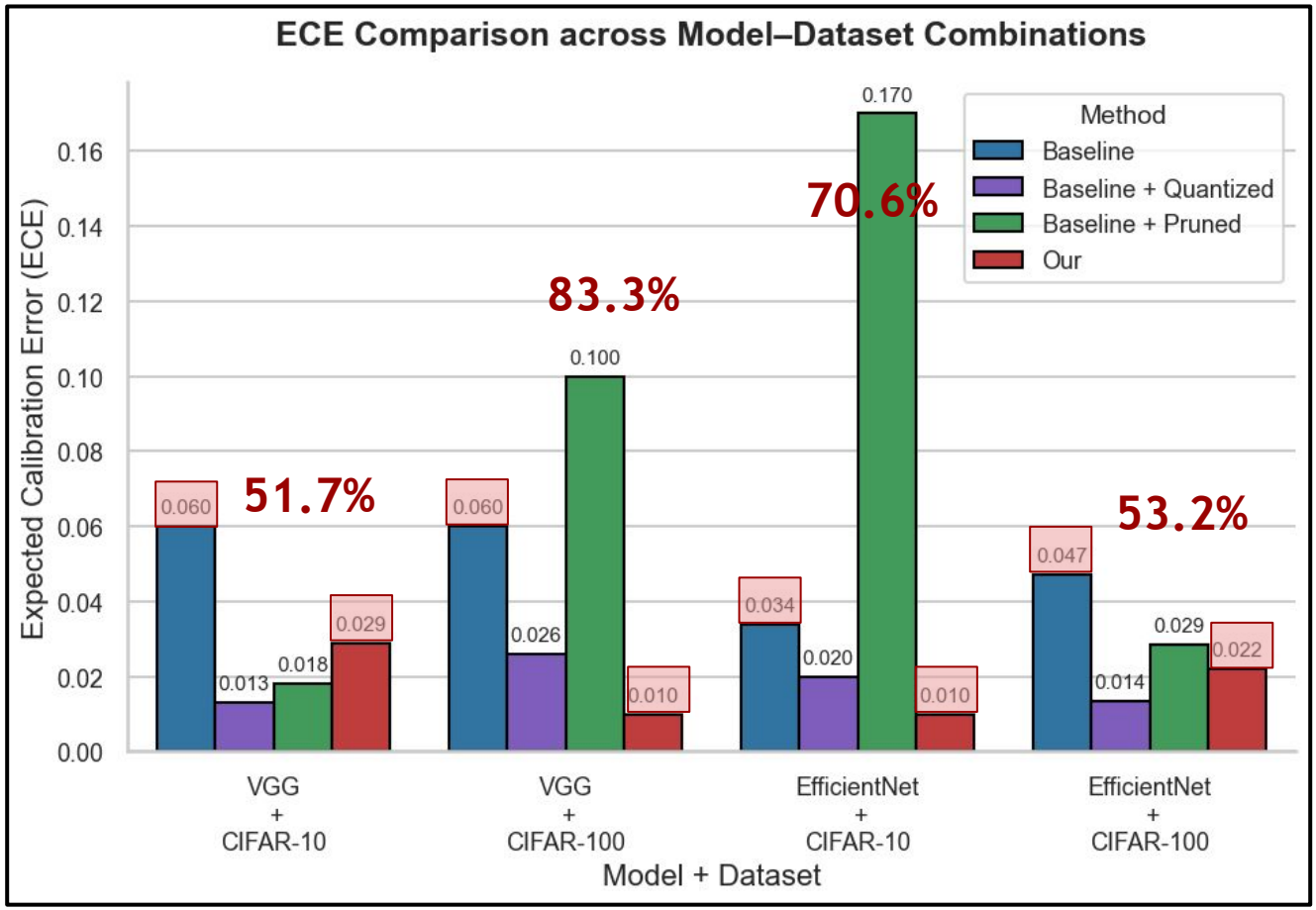
Expected Calibration Error (ECE)

↓ ECE → more confident and trustworthy predictions



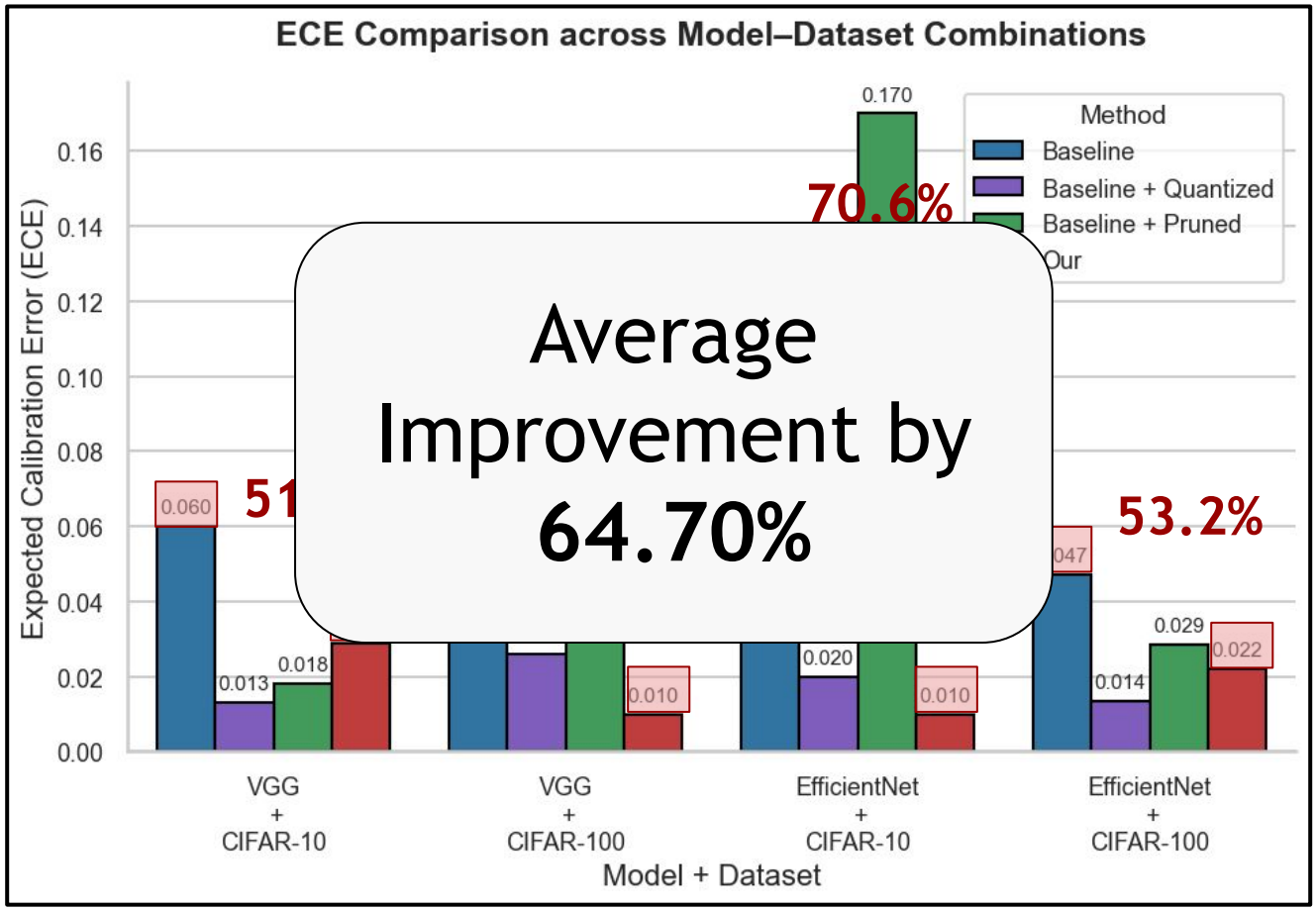
Expected Calibration Error (ECE)

↓ ECE → more confident and trustworthy predictions



Expected Calibration Error (ECE)

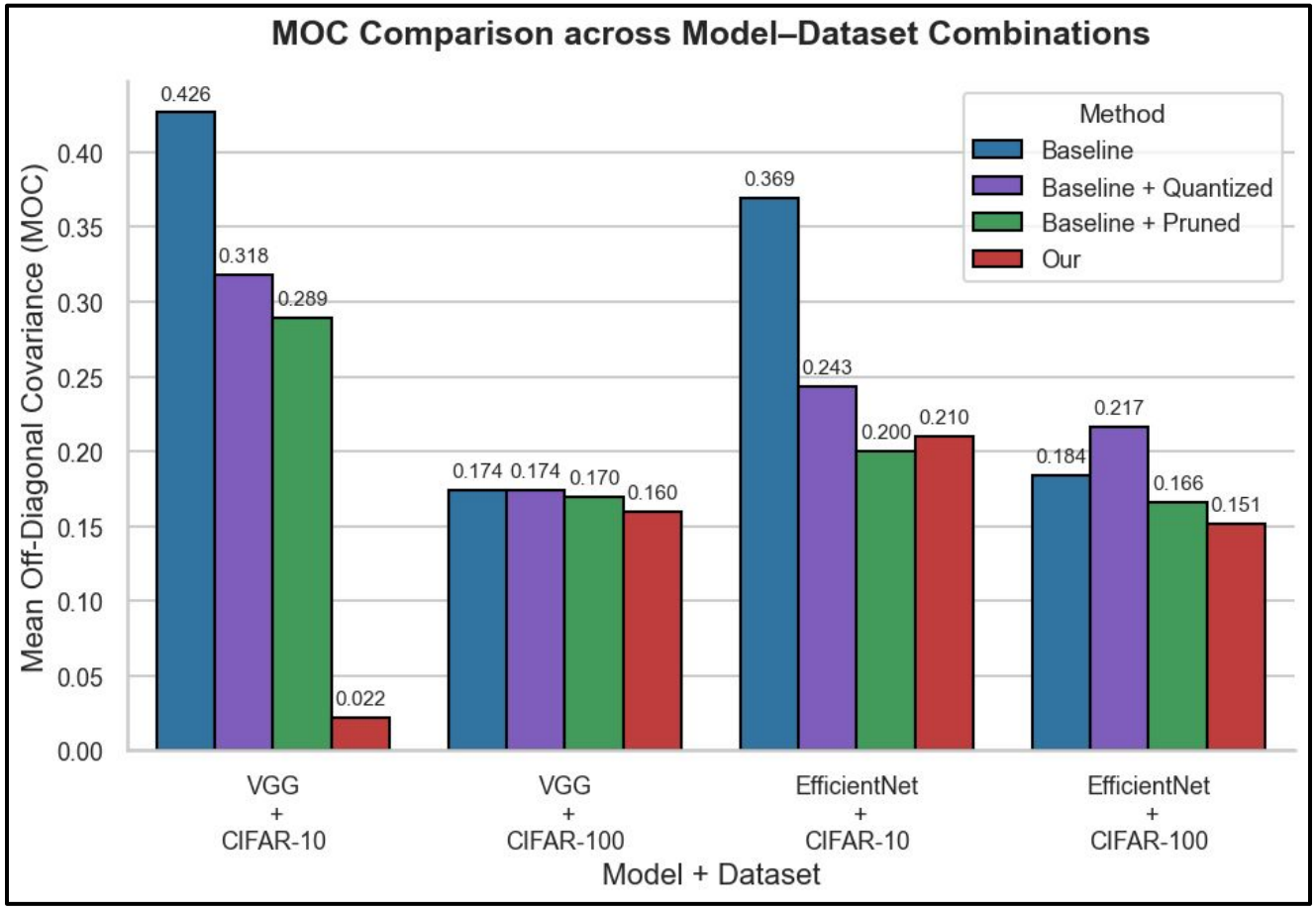
↓ ECE → more confident and trustworthy predictions



Average Improvement by **64.70%**

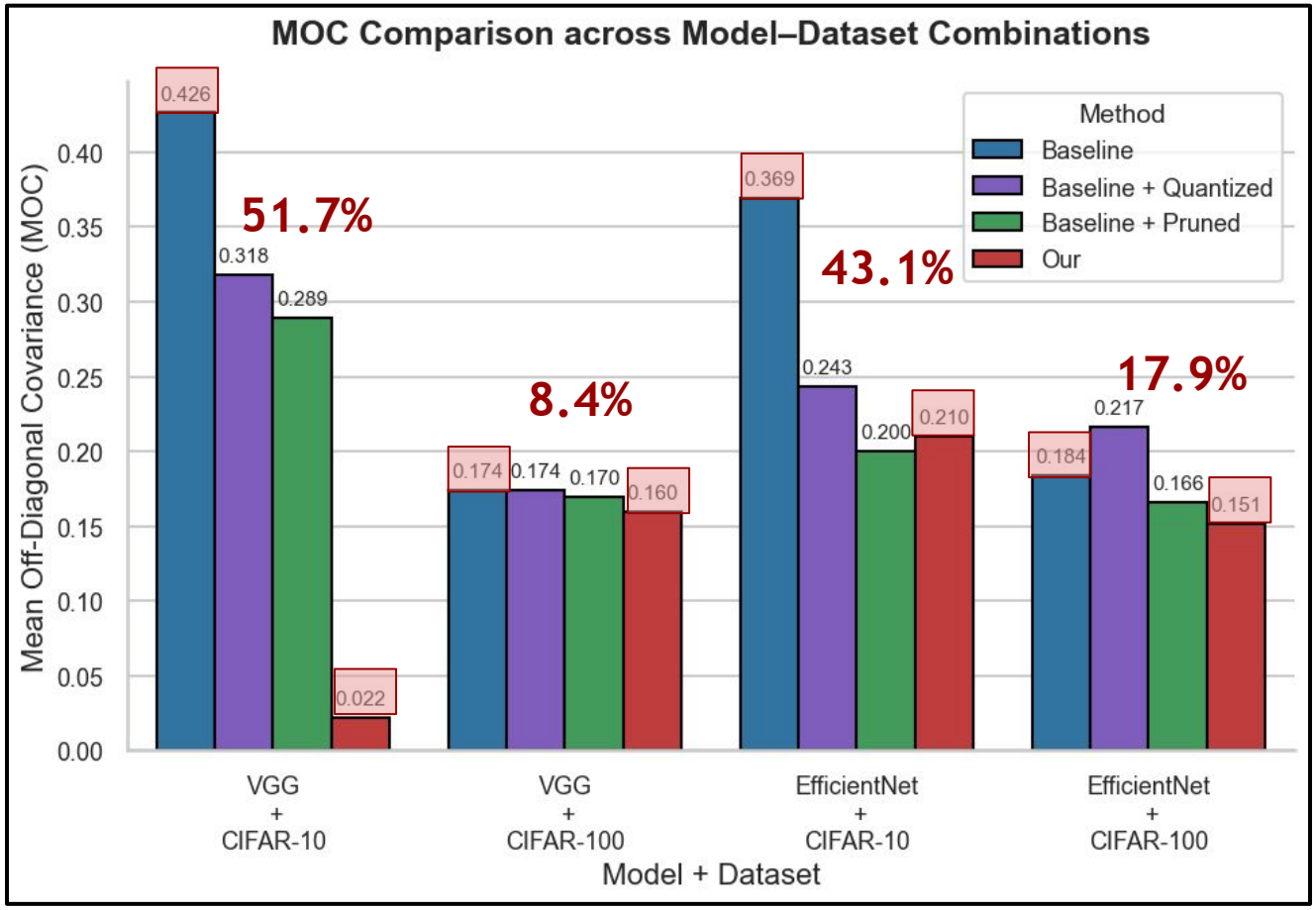
Mean Off-Diagonal Covariance (MOC)

↓ MOC → less redundancy among feature channels



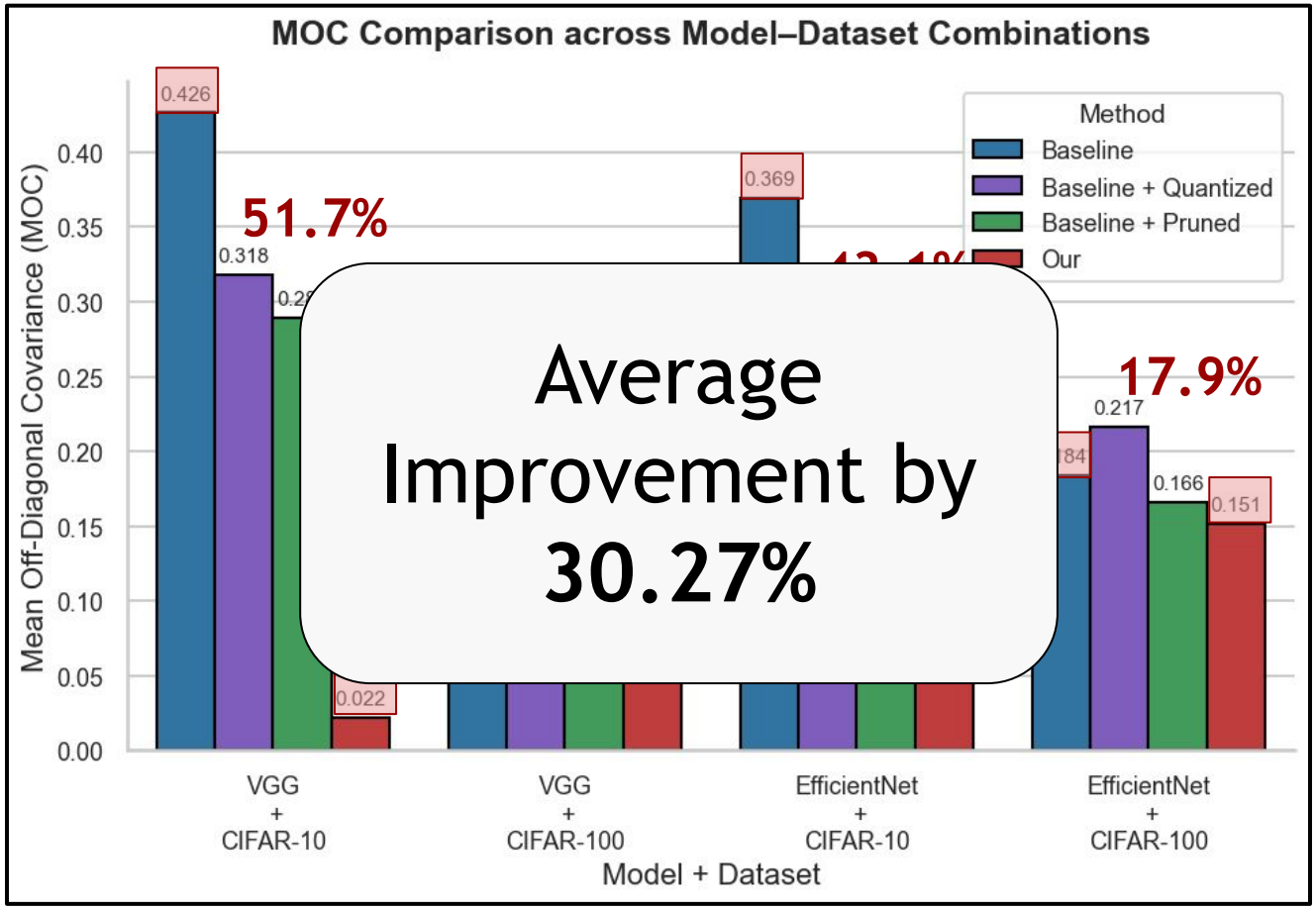
Mean Off-Diagonal Covariance (MOC)

↓ MOC → less redundancy among feature channels



Mean Off-Diagonal Covariance (MOC)

↓ MOC → less redundancy among feature channels



Average Improvement by **30.27%**

Test Accuracy

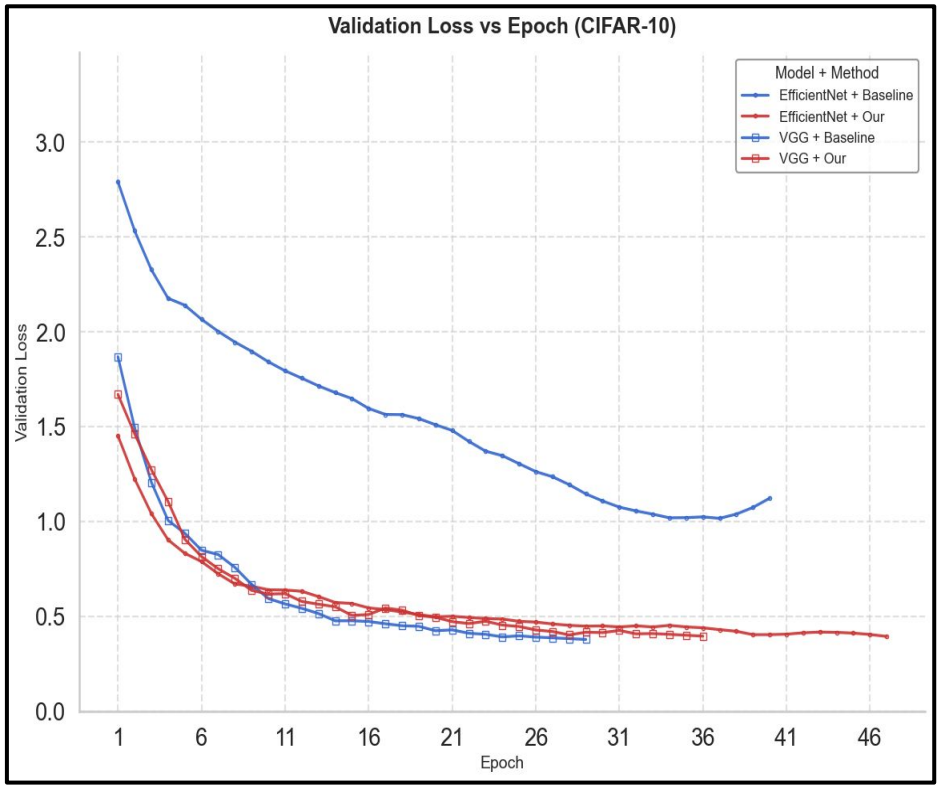
↑ Accuracy → better classification

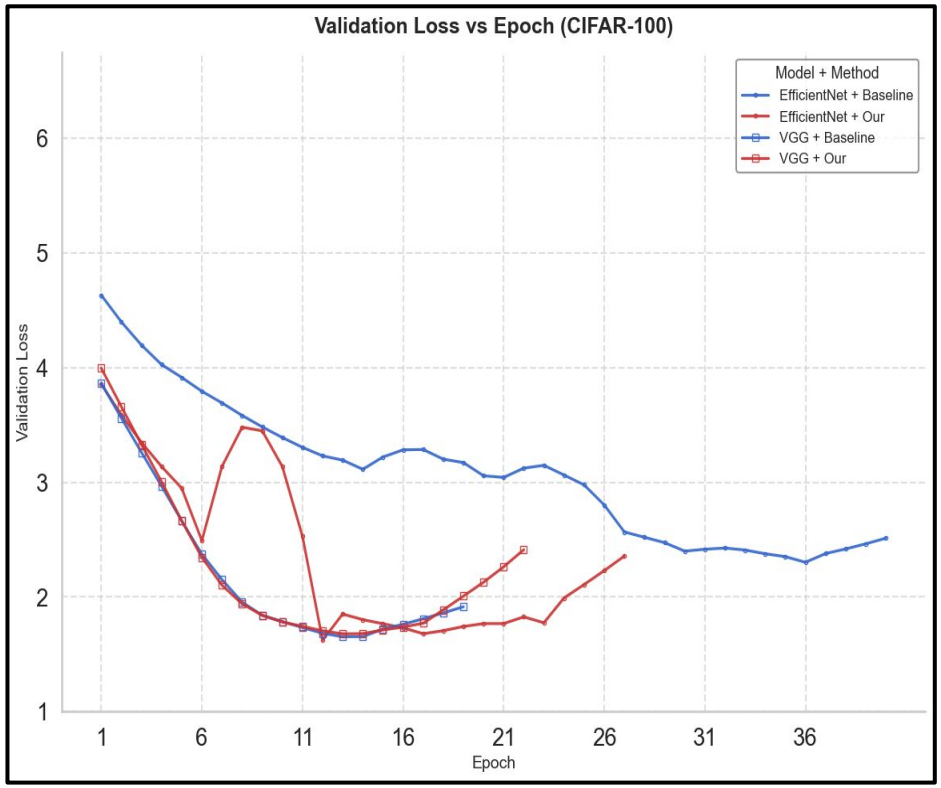
		VGG			
		Baseline	Baseline + Quantized	Baseline + Pruned	Our
CIFAR-10	Top-1%	76.14	86.29	89.29	88.03
	Top-5%	97.86	99.45	99.17	99.46
CIFAR-100	Top-1%	48.59	42.75	51.47	53.43
	Top-5%	78.6	73.93	81.45	83.75

		EfficientNet			
		Baseline	Baseline + Quantized	Baseline + Pruned	Our
CIFAR-10	Top-1%	43.41	10.02	83.54	86.68
	Top-5%	91.03	49.98	99.17	99.49
CIFAR-100	Top-1%	40.19	4.16	52.29	53.69
	Top-5%	71.64	11.77	82.24	83.11

Validation Loss

↓ Loss → better fit to true label distribution





Contribution

Literature Review:

1. *Ajith:* Read 2 papers - Suggested activation decorrelation loss
2. *Mrudani:* Read 2 papers - Suggested soft orthogonality loss
3. *Parth:* Read 2 papers - Suggested pruning and quantization aware losses

Training & Results:

1. *Ajith:* Quantized VGG and EfficientNet for both datasets. Applied our approach to EfficientNet CIFAR 100 dataset. Visualized an ensemble graph for ECE and Validation Loss.
2. *Mrudani:* Pruned VGG and EfficientNet for both datasets. Applied our approach to EfficientNet for CIFAR 10 dataset. Visualized an ensemble graph for train-test gap and MOC.
3. *Parth:* Baseline VGG and EfficientNet on both datasets. Applied our approach to VGG for both datasets. Visualized tables for top 1 & top 5 accuracy.

Thank You